



BSc (Hons) in Information Technology
Year 3 - Semester 2 (2026)
IT3060 - Human Computer Interaction

Lab Sheet 3

Submitted by:

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Group: 2.1

Activity 1: Review User Requirements

1.1 Pain Points Analyzed

Pain Point	Evidence	User Need
Limited personalization/ generic plans	5 out of 5 participants; Personalization Gap theme	Personalized plans based on individual schedule, fitness level, and goals
Cumbersome workout logging	High Friction Tracking theme; Onboarding process abandoned by 68% of participants	Easy tracking through fast logging (food recognition via camera, one-click workout logging)
Limited progress visualization	Quotes on dashboard clarity	Waning Motivation theme Dashboard with visual progress tracking and streaks
Lack of motivation due to social isolation	Social Isolation theme	Privacy preserving social circles for users and community challenges
Poor onboarding experience	App Analytics (68% onboarding drop-off)	Fast and efficient onboarding process with generation of personalized program right away

1.2 Key User Needs for New Features

- **AI-powered workout plans** - workout plans should automatically adapt to the user's timetable and physical condition, including injuries, and be easily recreated and rescheduled.
- **Social community** - private, customizable circles and challenges to create accountability without exposing users who value privacy.
- **Nutrition tracking** - log entries should be quick and painless, even better – camera-based, which will eliminate the only friction point mentioned.

1.3 Prioritized User Requirements

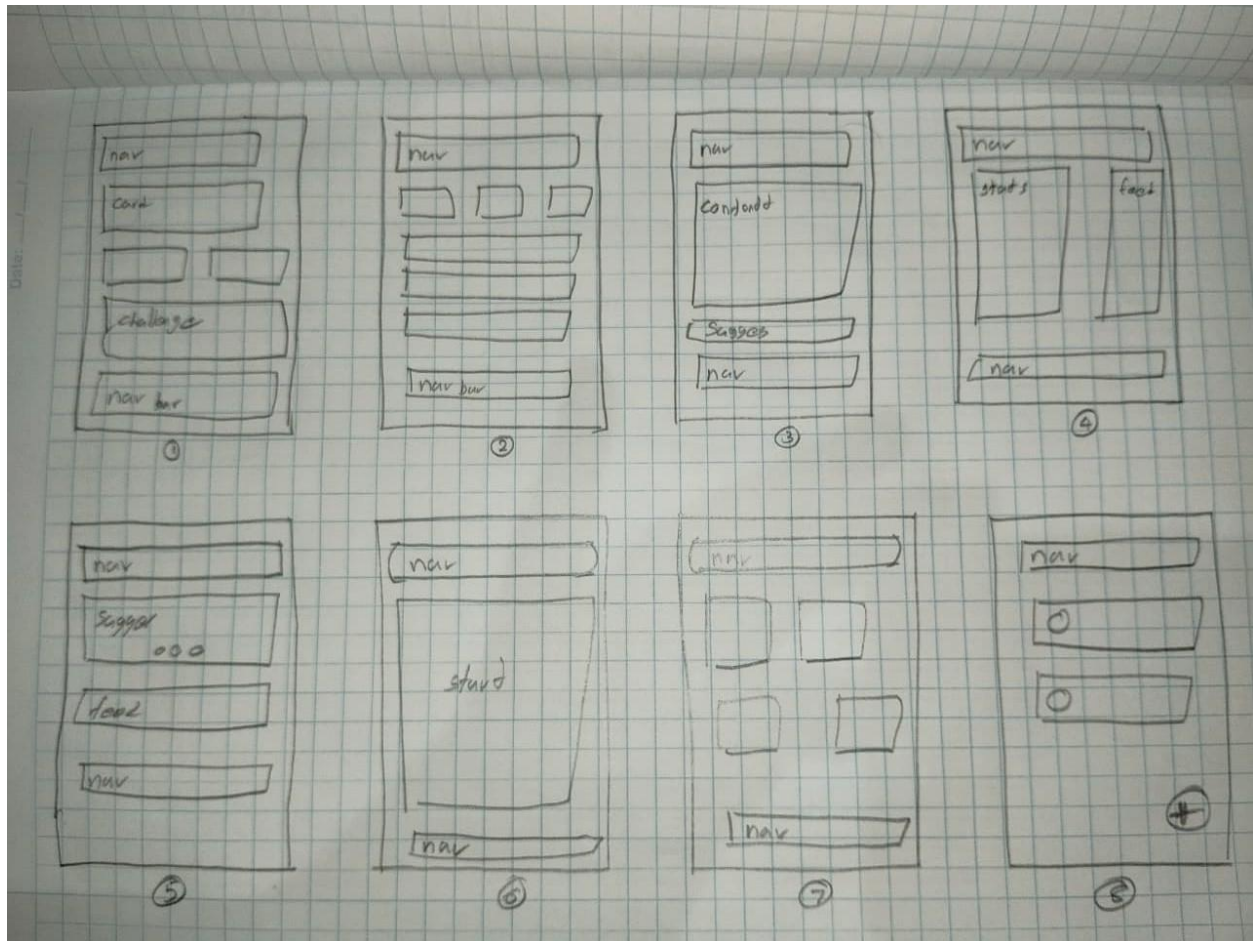
Requirement	Priority	Core User Flow
Provide an adaptive AI workout schedule for first use	High	Home Dashboard / Workout Planner
Take pictures of your meals and have them automatically identified	High	Nutrition Logger

Present a dashboard overview that gives you your day's plan, streak, and quick actions	High	Home Dashboard
Provide an option to create a social circle/challenges	High	Community Feed
Give progress trends, streaks and badges	Medium	Progress Tracking
Give users an option to reschedule/recreate missed workout	Medium	Workout Planner
Send reminders for both workouts and logging	Medium	Home Dashboard / Notifications
Give trainers an option to view and customize a client's AI plan	Medium	Workout Planner

1.4 Core User Flows Selected for Design

- **Home Dashboard** - starting screen that shows the 'AI Daily Flow' card, streaks, progress snapshot and shortcuts to other flows.
- **AI Workout Planner** - screen where an individual creates, edits and adjusts their adaptive workout plan using drag-and-drop functionality.
- **Progress Tracking** - graphs, streaks, badges and weekly AI insights.
- **Community Feed** - circles, challenges and privacy settings for social media sharing.
- **Nutrition Logger** - camera based nutrition log with a manual alternative and daily macros summary.

Activity 2: Crazy 8 Sketching



2.1 Notes on the Eight Concepts

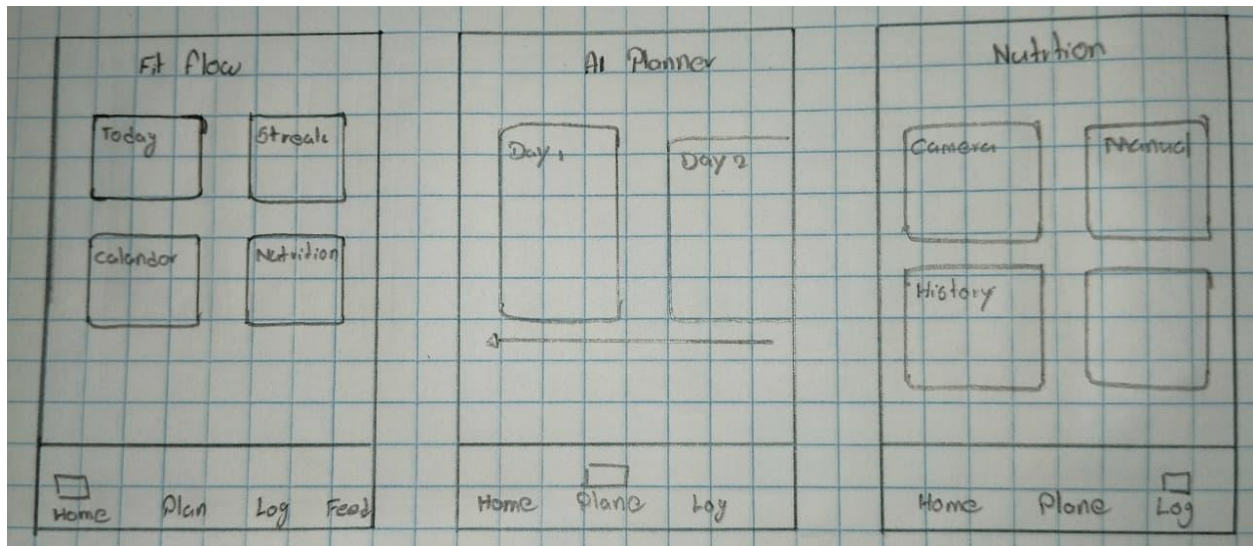
- **1 - Home Dashboard:** a large 'Today's Plan' card tops the screen with a clear "Start Now" button, so the AI recommendation gets acted on right away.
- **4 - Home Dashboard:** just like Idea 1, except the streak/ calories/ badges row below becomes a swipe able carousel to test whether swiping works as well as a fixed row while conserving vertical space.
- **2 - Home Dashboard:** structural navigation with a top tab strip (Home/ Plan/ Log/ Feed/ Profile).
- **5 - Home Dashboard:** same as Idea 2 except the navigation goes into a bottom tab strip compare reachability and visibility of top tabs against the bottom tabs.
- **3 - AI Workout Planner:** a conversational style dialogue that allows the AI to justify its recommendations for the week before presenting the plan.

- **6 - AI Workout Planner:** a full week plan view in card list format with drag and drop functionality and explicit 'Accept' action.
 - **7 - Community Feed:** The social feed and active challenge are moved way out front on the app to see if community content should have more importance than a secondary tab.
 - **8 - Nutrition Logger:** The 'scan meal' button is brought forward in its own tile on the Nutrition page that can be accessed straight from the Home Dashboard.
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- **Key findings:** Using a large 'Today's Plan' / AI card helped personalize and made it more prominent, but demoted community and nutrition further down on the screen.
 - **Top tabs versus bottom nav** - The bottom tab bar performed better as one-hand navigation, with a thumb reaching all five flows.
 - **Conversational Planner** - Helped build credibility for the AI's reasoning but took longer to scan than a simple list when the user simply wanted to start exercising.
 - **Nutrition Tile** - Addressed the problem of 'cumbersome workout/meal logging' stated in Activity 1, as now it is done in one tap.
 - **Carried forward** - Three most promising directions Card/Tile Grid Dashboard, Conversational AI-first planner, and Hybrid Quick Action Grid were carried to Activity 3 as fully developed variants.

Activity 3: Produce Multiple Interface Variation

3.1 Card/Tile Grid Dashboard

- **Home Dashboard:** 2 column grids of cards Today's Plan, streak, calories, and a Nutrition tile are all equally weighted tiles.
- **AI Workout Planner:** Plan days presented as a scrollable row of cards rather than a vertical list.
- **Nutrition Logger:** Camera button is one tile amongst others rather than a focal point on the screen.
- **Trade off:** Very scannable and modular, but nothing is emphasized, making the AI recommendation difficult to find within equally weighted tiles.



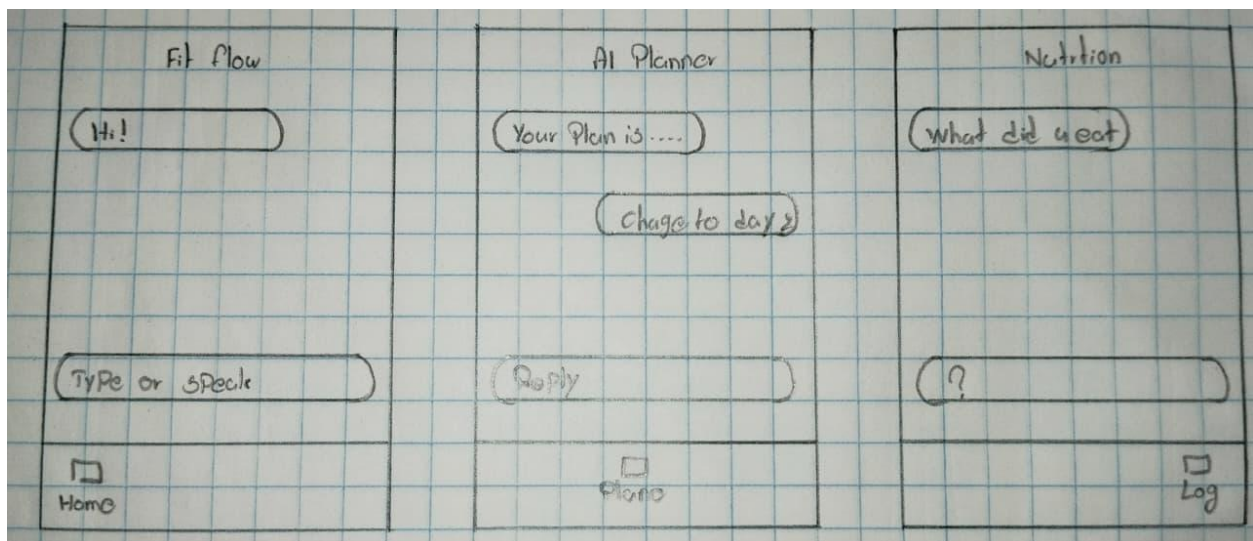
3.2 Conversational AI-First

Home Dashboard: Chat-based AI greeting that appears as soon as the app is opened rather than dashboard with quickly scannable statistics hidden in the drawer.

AI Workout Planner: Fully conversational experience where the AI proposes a plan and the user accepts/edits it through chat rather than scrolling through cards list.

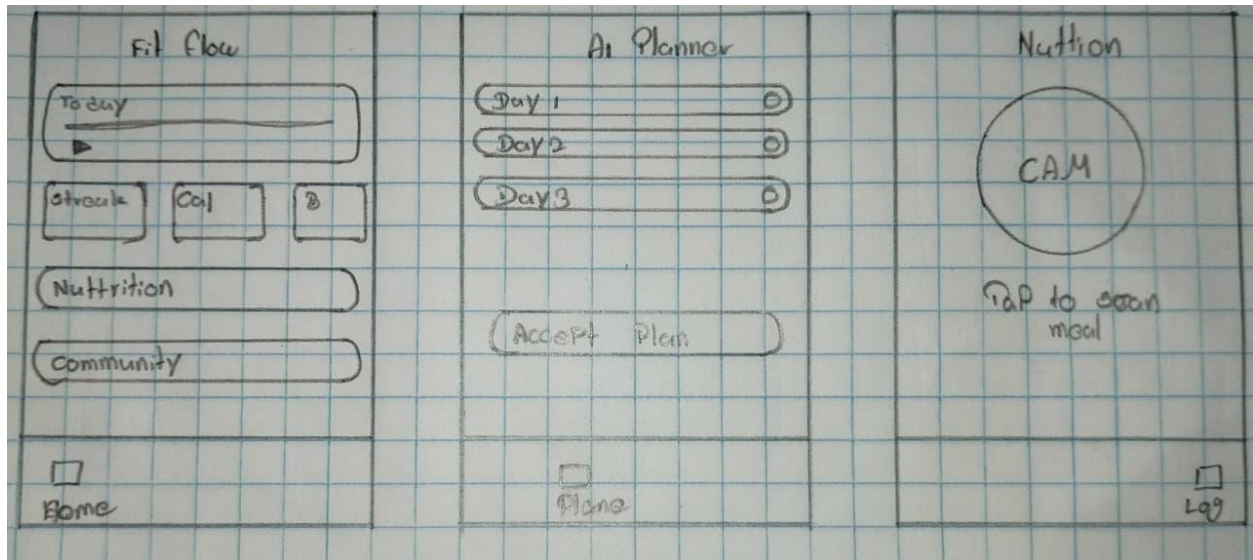
Nutrition Logger: Meals entered in chat format by telling them to the AI rather than taking picture with the camera.

Trade off: Most innovative and personalized (highest innovation score) of all possible paths but takes more time to perform repeated routine actions and harder to develop consistently.



3.3 Hybrid Quick-Action Grid

- **Home Dashboard:** A single Today's Plan card with the obvious Start Now button and a See full plan link, a small stats bar, a nutrition tile with the Log Meal button and a Community section each with a single action.
- **AI Workout Planner:** Full week displayed in a drag & drop editable list of day cards with a single Accept Plan button leading back to the Home Dashboard.
- **Nutrition Logger:** Large camera-first button takes over the whole screen, with a small 'enter manually' link as a fallback.
- **Navigation:** Persistent bottom tab bar (Home/ Plan/ Log/ Feed/ Profile) displayed on all five core flows.
- **The trade off:** The interface remains consistent and speedy to use on a daily basis; less innovative than conversational approach, but the most usable and the easiest to implement on selected React Native/ TensorFlow Lite stack.



Activity 4: Select the Best Desing Using Justification

4.1 Decision Matrix (1 = weak, 5 = strong)

Criteria	Card/Tile Grid Dashboard	Conversational AI First	Hybrid Quick Action Grid
Usability	4	3	5
Visual appeal	4	4	4
Alignment with user needs	4	3	4
Innovation	3	5	3

Feasibility (React Native / TensorFlow Lite stack)	5	3	5
Total (out of 25)	20	18	21

4.2 Selected Design Direction

Hybrid Quick Action Grid - Combination Grid of Quick Actions (Bottom Navigation Bar) was chosen due to its high score (21/25) with a high level of usability and feasibility and competitive scores in attractiveness and user oriented.

4.3 Justification

The hybrid quick action network addresses all five of the issues identified in the user research collected in Table 1 of Activity 1. The 68% drop in engagement, as well as the high-friction tracking issue, was largely due to how difficult it was to track a workout or meal. To address this issue, there is now only one button found on the original dashboard, and a separate nutrition tile with a quick action to Log Meals, meaning users can access both basic actions with a single click from the main page.

Regarding the personalization gap, it is addressed by the fact that the Today's Plan card, a suggestion made by AI, is still the primary focus of the dashboard, and a user can click See Full Plan to get more information about it if they want. Unlike the first concept of conversational AI, personalization will be prominently displayed even if interaction with the chat is not required.

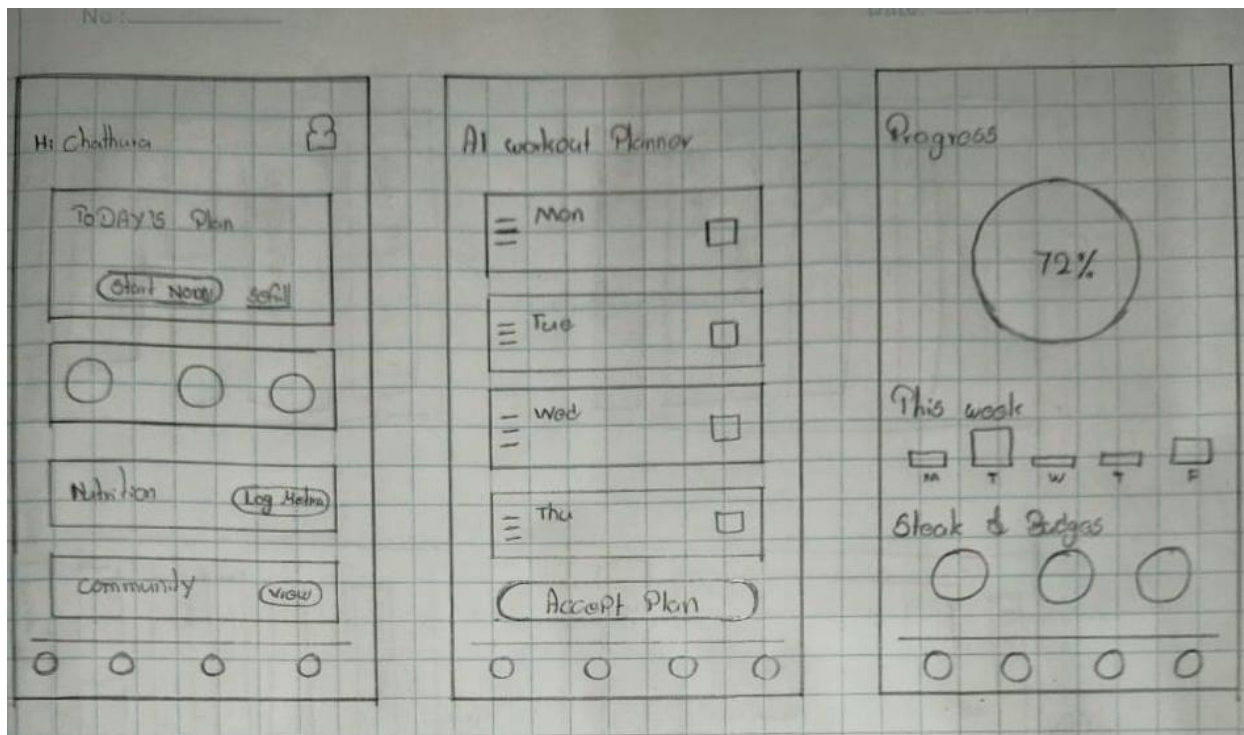
The themes of Waning Motivation and Social Isolation are served by a separate progress section below the design card that displays bars, calories, and badges, while the theme of Community Excitement is served by providing engagement through personally selected circles, as opposed to an activity stream. This is consistent with the privacy-oriented needs listed in Table 1 and Section 1.2.

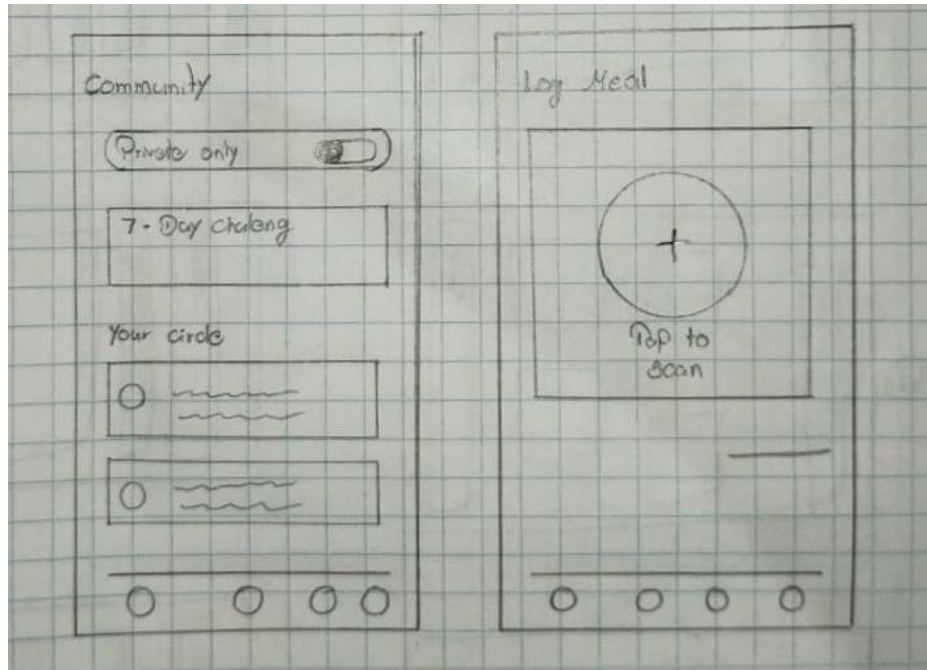
According to the decision matrix, the Hybrid Quick-Action Grid received the highest ratings for Usability (5) and Feasibility (5) because it includes simple React Native components that are easy to use. On the other hand, the conversational AI-first design concept includes another layer of conversation in terms of interaction design. Although the Hybrid Quick-Action Grid received a relatively low rating for innovation (3), it was considered an acceptable compromise given the primary goal of the redesign process, which was to facilitate reduced onboarding friction, daily tracking, and ultimately improved user retention. As a result, the Hybrid Quick-Action Grid concept was ultimately selected for further development in Activities 5 and 6.

Activity 5: Create Paper and Digital Wireframes

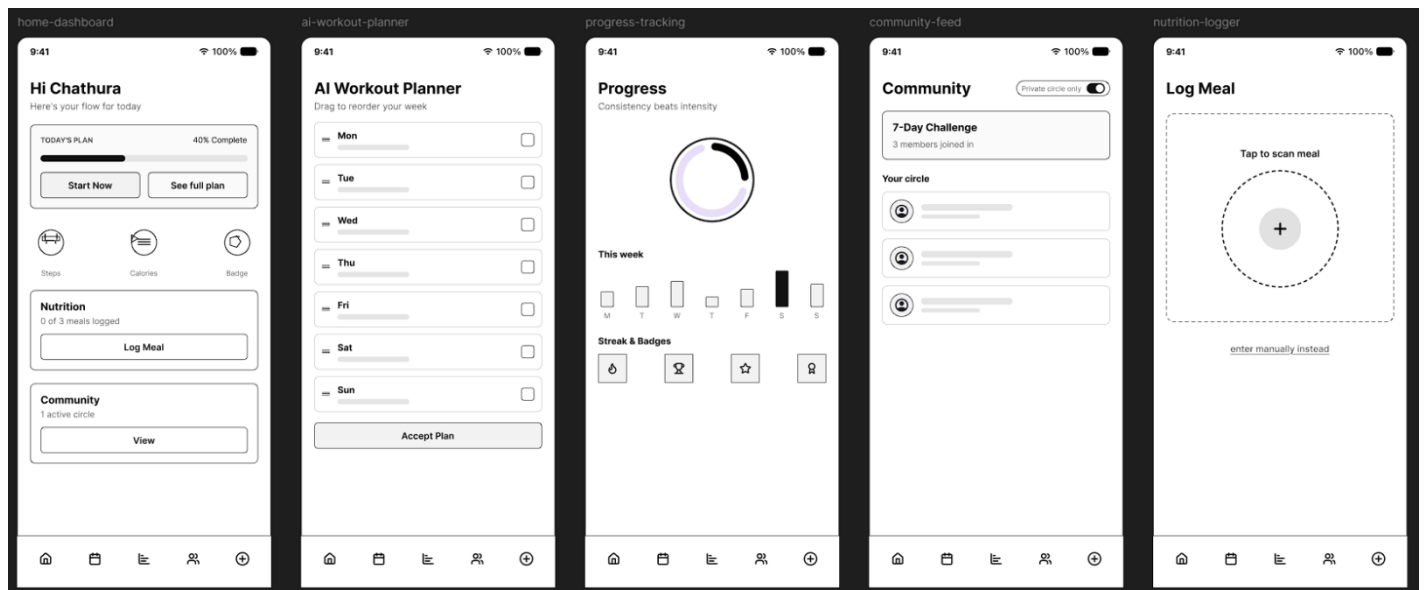
5.1 Paper Wireframes

Before exploring the use of digital tools, low-fidelity paper wireframes were manually prepared for each of the five main user flow scenarios, as specified in Section 1.4, according to the hybrid quick-action grid method chosen in Activity 4. These notes focused on the layout and hierarchy of the content, rather than on visual design, and maintained the five-tab bottom navigation (Home/Plan/Login/Feed/Profile) from the Crazy 8 notes stage.





5.2 Digital Low-Fidelity Wireframes



Key elements:

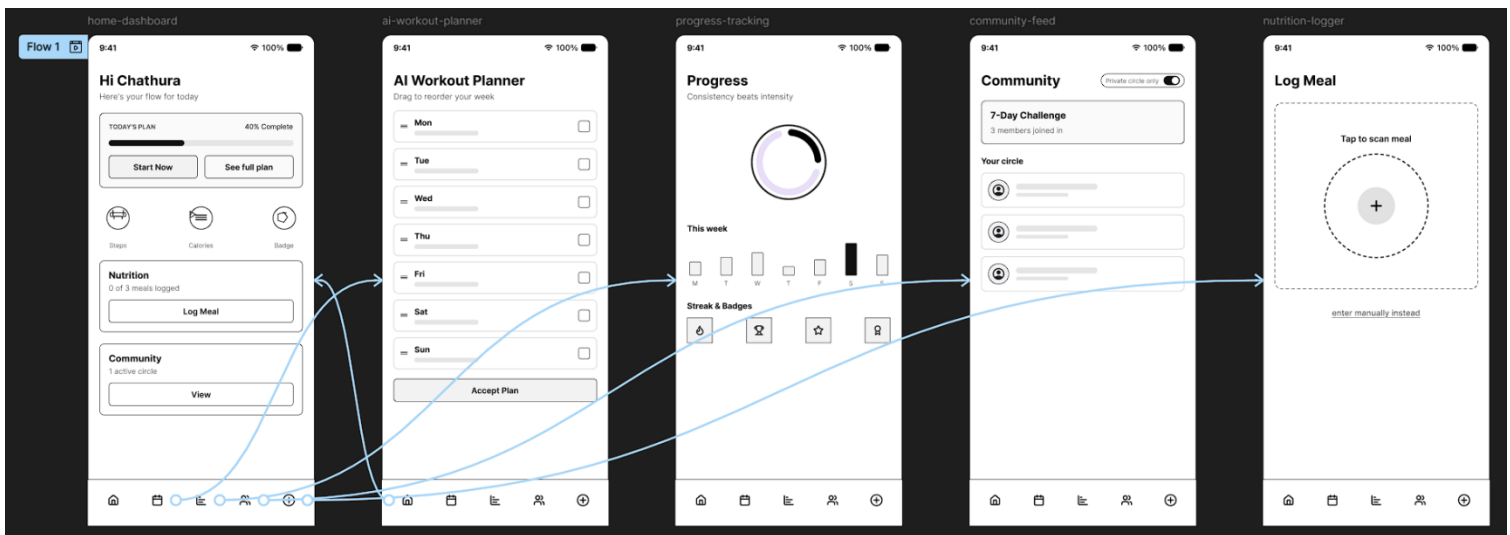
- **Home Dashboard** - Today's Plan card has a Start Now button and a View Full Plan link, and a small streak/calorie/badge stats bar with single-action

nutrition and community tiles below it, targeting the personalization gap and high-friction tracking issues.

- **AI Workout Planner** - The entire week is displayed as a vertical list of draggable day cards with a single Accept Plan button, allowing you to reorder it without switching to another view, similar to the 3.3 description.
- **Progress Tracking** - Uses the same ring component as the Home Dashboard hero card for a consistent look, combined with a simple bar chart for the week and a row of badges to target the Waning Motivation issue.
- **Community Feed** - To address the privacy first need (NFR4) derived from Participant 5's feedback, a private circle toggle is displayed above the view and not a public feed.
- **Nutrition Logger** - Has a prominent single camera first scan button, and the 'Enter manually' link is used only as a fallback option, addressing the preference from FR3 and a pilot test.

Activity 6: Build a Clickable Low Fidelity Prototype

6.1 Prototype Overview



Prototype link: <https://www.figma.com/design/okKCCGZkECwpGIDEObtVxi/HCI-lab-3-UI?node-id=0-1&t=pHyXuifVq7nHNYVT-1>

6.2 Linked Flows

From	Trigger	To	Purpose
Home Dashboard	'Start Now' button	AI Workout Planner	Pathway for transition from the single focus call to action to the present plan
Home Dashboard	Nutrition tile	Nutrition Logger	Direct pathway to record meals without moving away from the main flow
Home Dashboard	'See full plan' link	AI Workout Planner (full view)	Alternative pathway for those who wish to view the entire week
Home Dashboard	'Community' link/ Feed nav icon	Community Feed	Option to access the social network, aligning with the privacy research
Any screen	Bottom nav bar	Home/ Plan/ Log/ Feed/ Profile	Global navigation throughout all major flows
Workout Planner	'Accept Plan'	Home Dashboard	Returns user to the dashboard with plan confirmation

6.3 Annotations / Key Design Decisions

- **Home Dashboard** - “Start Now is the main CTA for the AI plan, and the Nutrition tile gives a one-touch link to logging.”
- **Workout Planner using AI** - “Handle drag lets you rearrange days; Accept Plan takes you back to Home Dashboard.”
- **Nutrition Logging** - “Camera button makes data entry automatic; manual entry is there only as a backup option.”
- **Community Feed** - “Circle is private, not a timeline feature as per privacy research; the switch is at the top of the feed.”
- **Progress Tracking** - “Ring has been used again from the Home Dashboard here for consistency.”